# FINAL PROJECT REPORT

**Project Title :**

*Ride Cancellation Prediction*

**Course :**

*UYG338 – Software Project Management*

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# *Abstract*


## *Project Purpose*

*This project focuses on predicting ride cancellation probabilities using machine learning techniques to improve operational efficiency in ride-hailing platforms. Ride cancellations may cause customer dissatisfaction, increased waiting times, and inefficient resource management. The aim of this project is to identify possible cancellations in advance and support better decision-making processes.*

## *Technologies Used*

*The dataset includes ride-related features such as request time, location, driver behavior, customer activity, and historical cancellation records. Python was used as the main programming language, while Pandas, NumPy, and Scikit-learn libraries were utilized for data preprocessing and model development. Logistic Regression, Decision Tree, and Random Forest algorithms were implemented and compared to evaluate prediction performance.*

## *Main Results*

*Unlike a standard academic study, this project was designed as an operational and business-oriented prototype for review by business development specialists. Therefore, the application was intentionally kept in a local environment instead of being deployed globally. The final model achieved meaningful accuracy in predicting ride cancellations and demonstrated the practical value of predictive analytics in transportation systems.*

# *Introduction*

## *Importance of the Project*

*Ride cancellation is one of the major operational problems in ride-hailing platforms. Frequent cancellations may lead to customer dissatisfaction, increased waiting times, inefficient driver allocation, and financial losses for companies. Predicting cancellations before they occur can help transportation platforms optimize their operations and improve service quality. With the growing use of data-driven technologies, machine learning provides effective solutions for analyzing user behavior and forecasting potential ride outcomes.*

## *Objectives*

*The main objectives of this project are:*

- *To analyze ride-related data and identify factors affecting cancellations*
- *To develop machine learning models capable of predicting ride cancellation probabilities*
- *To compare different algorithms such as Logistic Regression, Decision Tree, and Random Forest*
- *To create a simple operational prototype for testing prediction results*
- *To support business-oriented decision-making processes through predictive analytics*

## *Scope*

*This project includes data preprocessing, exploratory data analysis, feature engineering, machine learning model development, evaluation, and a basic local deployment interface using Streamlit. The system was designed as an operational prototype for demonstration and business evaluation purposes. Global deployment, real-time integration, and large-scale production infrastructure are outside the scope of this project.*

# *Project Management*

## *Project Timeline (Weekly Plan)*

***Week 1 – Problem Definition and Business Understanding***

*During the first week, the business problem and the operational impact of ride cancellations were analyzed. Stakeholders and project objectives were identified, while the dataset structure and project scope were examined.*

***Deliverables:*** *Problem statement, project scope document, business value explanation.*

***Week 2 – Data Exploration and Understanding***

*The dataset was loaded and inspected. Missing values and inconsistent records were handled, and exploratory data analysis (EDA) was performed to understand cancellation patterns and feature distributions.*

***Deliverables:*** *Cleaned dataset, EDA visualizations, initial insights report.*

***Week 3 – Feature Engineering and Data Preprocessing***

*In this phase, categorical variables were encoded, numerical features were scaled, and additional features were created to improve model performance. The dataset was also split into training and testing sets in preparation for model development. This step ensured that the data was structured properly and suitable for machine learning algorithms.*

***Deliverables:*** *Initial feature set, preprocessing pipeline, train-test datasets*

***Week 4 – Model Development (First Iteration)***

*In this phase, multiple machine learning models including Logistic Regression, Decision Tree, and Random Forest were trained using the prepared dataset. Initial model evaluation was performed to compare their performance and identify the most suitable approach for the problem.*

***Deliverables:*** *First trained models, initial model comparison results.*

***Week 5 – Data Leakage Detection and Iteration Phase***

*During model evaluation, a data leakage issue was identified, which indicated that some features were unintentionally influencing the target variable and leading to unreliable performance results. As a result, the feature engineering process was revisited and the dataset was restructured to eliminate leakage risks. Problematic features were removed or adjusted, and the preprocessing pipeline was updated to ensure data integrity.*

***Deliverables:*** *Data leakage analysis report, revised feature engineering strategy, updated dataset structure.*

***Week 6 – Model Development (Second Iteration)***

*After correcting the data leakage issue, machine learning models were retrained using the updated feature set. Logistic Regression, Decision Tree, and Random Forest models were re-evaluated, and their performance was compared again. This iteration resulted in more reliable and realistic model outcomes.*

***Deliverables:*** *Retrained models, improved performance comparison results.*

***Week 7 – Deployment***

*A local deployment interface was developed using Streamlit. The deployed prototype was tested to validate prediction outputs and system functionality.*

***Deliverables:*** *Locally deployed operational prototype.*

***Week 8 – Final Documentation and Portfolio***

*The final project report, technical documentation, and GitHub repository organization were completed. The project was prepared as a portfolio-ready academic and operational case study.*

***Deliverables:*** *Final report, technical documentation, GitHub portfolio repository.*

## *Task Distribution*

*This project was completed individually by me, Zehra Buse Tüfekçi. Therefore, all roles and responsibilities were handled by a single team member throughout the entire development lifecycle. Although different roles were defined for project management purposes, all of them were executed by me in a structured and organized manner.*

*The responsibilities were distributed as follows: I acted as the Project Manager, responsible for planning the project timeline, defining milestones, and managing the overall workflow. As the ML Developer, I handled data preprocessing, feature engineering, model development, and evaluation processes. In the role of Backend Developer, I implemented the core logic of the system and ensured model integration. As the Frontend Developer, I developed the Streamlit interface for interactive prediction. I also took the role of Tester / QA, performing debugging, validation, and model performance checks. Finally, as the Documentation Specialist, I prepared all technical documentation, reports, and the final project submission.*

*This structure reflects a multi-role approach within a single-person project, ensuring that all aspects of software development and project management were fully covered.*

## *Methodology – Scrum (Agile)*

*For this project, the Scrum methodology (Agile framework) was selected. The main reason for choosing Scrum is that it provides a flexible, iterative, and well-structured workflow, which is especially suitable for managing complex machine learning projects with evolving requirements.*

*Although Scrum is typically used in team-based environments, it was adapted for this project as a single-person development approach. Jira was used as the project management tool to organize tasks, create a structured backlog, and track progress through weekly sprints. This helped maintain discipline and clarity throughout the development process.*

*Scrum was preferred over Waterfall because Waterfall follows a linear structure and does not easily support iteration. In this project, multiple iterations were required, especially after detecting data leakage and revising the feature engineering and model development phases. Scrum allowed these revisions to be handled efficiently without disrupting the overall project flow.*

*As a result, Scrum made it easier to manage tasks, monitor progress, and continuously improve the model in an organized way, even in an individual project setting.*

# *Risk Management*

## *Identified risks*

*The following risks were identified for the*
***Ride Cancellation Prediction Project:***

- ***Data Quality Risk:*** *The dataset may include missing values, inconsistent entries, or incorrect formatting, which can negatively affect model performance.*

- ***Model Performance Risk:*** *Machine learning models may not achieve the expected level of accuracy or may perform poorly on unseen data.*

- ***Overfitting Risk:*** *The model may learn patterns too closely from the training data and fail to generalize to new data.*

- ***Deployment Risk:*** *The Streamlit-based application may encounter issues due to dependency conflicts or environment setup problems.*

- ***Time Management Risk:*** *Project stages such as preprocessing, modeling, and documentation may take longer than planned.*

- ***Dependency Compatibility Risk:*** *Different versions of Python libraries may cause compatibility issues during development or deployment.*

## *Risk Analysis*

*Each risk was evaluated based on its likelihood and potential impact:*

- ***Data Quality Issues*** *were assessed as* ***high likelihood and high impact****, making it the most critical risk in the project.*

- ***Model Performance Issues*** *were evaluated as* ***medium likelihood with high impact****, as poor performance would directly affect prediction reliability.*

- ***Overfitting*** *was considered* ***medium likelihood and medium impact*** *due to its dependency on model selection and validation strategy.*

- ***Deployment Issues*** *were assessed as* ***medium likelihood and medium impact*** *because environment-related problems are common in local deployment setups.*

- ***Time Management Problems*** *were also evaluated as* ***medium likelihood and medium impact*** *due to iterative development and debugging phases*

- ***Dependency Conflicts*** *were considered* ***low likelihood but medium impact,*** *as they can usually be resolved with version control.*

***Overall,*** *data-related risks were identified as the most critical category because they directly influence the quality and reliability of the machine learning model.*

## *Mitigation Strategies*

*To reduce the impact of the identified risks, the following strategies were applied:*

***Data Quality Risk Mitigation:***

- *Data cleaning techniques were applied, including handling missing values, removing duplicates, and correcting inconsistent records to ensure dataset reliability.*

***Model Performance Risk Mitigation:***

- *Multiple machine learning models were tested, and cross-validation techniques were used to improve reliability and generalization performance.*

***Overfitting Risk Mitigation:***

- *Train-test splitting was applied, and model performance was monitored on unseen data to ensure proper generalization.*

***Deployment Risk Mitigation:***

- *All dependencies were clearly defined in a requirements file, and the deployment environment was tested in a clean setup to prevent runtime errors.*

***Time Management Risk Mitigation:***

- *The project was structured using iterative planning, and tasks were divided into smaller milestones following an Agile workflow.*

***Dependency Compatibility Mitigation:***

- *Consistent library versions were used, and all dependencies were documented to avoid version conflicts during development and deployment.*

# *Time and Resource Management*

## *Work Packages*

*The project was structured into sequential and iterative work packages based on the full development lifecycle:*

***Work Package 1:***
***Problem Definition and Business Understanding (Week 1)***

- *Identification of the business problem, stakeholders, and success metrics, along with defining the project scope and dataset understanding.*

***Work Package 2:***
***Data Exploration and Understanding (Week 2)***

- *Data loading, inspection, handling missing values, and performing exploratory data analysis (EDA) to understand cancellation patterns and feature behavior.*

***Work Package 3:***
***Feature Engineering and Data Preprocessing (Week 3)***

- *Encoding categorical variables, scaling numerical features, feature creation, and preparing train-test datasets for model development.*

***Work Package 4:***

***Model Development – First Iteration (Week 4)***

- *Training initial machine learning models (Logistic Regression, Decision Tree, Random Forest) and performing baseline evaluation and comparison.*

***Work Package 5:***

- ***Data Leakage Detection and Iteration (Week 5)***
- *Identification of data leakage issues during evaluation, revision of feature engineering approach, removal of problematic features, and restructuring of the dataset.*

***Work Package 6:***

***Model Development – Second Iteration (Week 6)***

- *Retraining models using the corrected dataset and re-evaluating performance to obtain more reliable results.*

***Work Package 7:***

***Deployment (Week 7)***

- *Development and testing of a local Streamlit interface for model deployment and prediction demonstration.*

***Work Package 8:***

***Final Documentation and Portfolio (Week 8)***

- *Preparation of the final report, technical documentation, and organization of the GitHub repository as a portfolio-ready project.*

## *Planned vs Actual Effort*

*The project was initially planned as a structured 60-hour effort; however, the actual process extended slightly due to iterative improvements and corrections.*

***Planned Effort:*** *60 hours across 7–8 weeks*
***Actual Effort:*** *Slightly higher than planned due to data leakage detection and model retraining phases*

***Main Reason for Deviation:*** *Iterative refinement of feature engineering and model development to ensure reliable results*

## *Tools Used*

*For project management and task tracking,* ***Jira*** *was selected as the primary tool. Jira was chosen because of its ease of use and its widespread adoption in industry, especially in* ***Scrum-based workflows,*** *which were applied in this project.*
*Even though this was a single-person project, Jira was used effectively to manage tasks through sprints, organize weekly work packages, and monitor progress in a structured manner.*

***Additional tools used in the project include:***

- *Python (core programming language)*
- *Pandas, NumPy, Scikit-learn (data processing and machine learning)*
- *Streamlit (local deployment interface)*
- *GitHub (version control and portfolio management)*
- *Local environment / Google Colab (development and testing)*

# *Github Repository*

## *Repository Link:*

*https://github.com/zehrabuse/ride-cancellation-prediction-ml*

## *Description:*

- *The GitHub repository was used as the central source code management platform throughout the project.*

- *All project development, including data preprocessing, feature engineering, model training, and deployment, was maintained in a single main branch without additional branching strategies.*

- *The development process followed a continuous improvement approach, where new features, fixes, and model updates were frequently committed in short intervals.*

- *Commit history reflects an iterative workflow, with regular updates made to refine the model, fix issues such as data leakage, and improve overall system performance.*

- *This approach ensured a simple but effective version control structure suitable for an individual academic project.*

- *GitHub also served as a portfolio platform to present the complete end-to-end machine learning pipeline in a structured and accessible way.*

# *Jira Project Board*

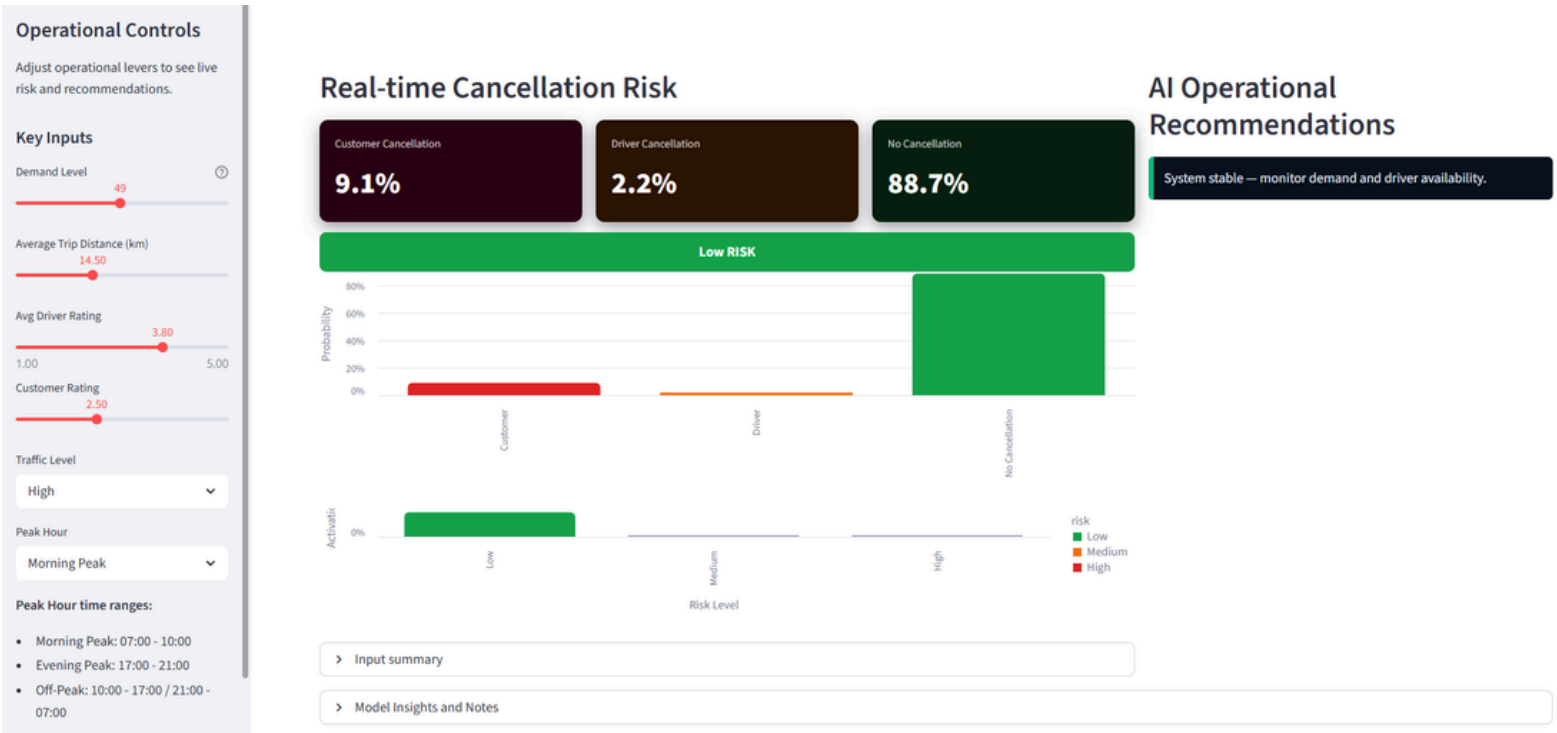
## *Jira Board Link:*

*https://zehrabusetufekci.atlassian.net/jira/software/projects/SCRUM/boards/1*

## *Description:*

- *Jira was used as a task tracking and project organization tool based on the Scrum framework.*
- *Tasks were managed through sprint-based planning, allowing structured weekly progress tracking.*
- *Issue types included tasks for data processing, model development, deployment, and documentation activities.*
- *Although the project was completed individually, Jira helped maintain discipline, organize workload, and monitor progress in a structured workflow.*

# *Project Demonstration (Local Deployment Evidence)*



## Operational Controls

Adjust operational levers to see live risk and recommendations.

### Key Inputs

Demand Level

49

Average Trip Distance (km)

14.50

Avg Driver Rating

3.40

Customer Rating

1.30

Traffic Level

Severe

Peak Hour

Evening Peak

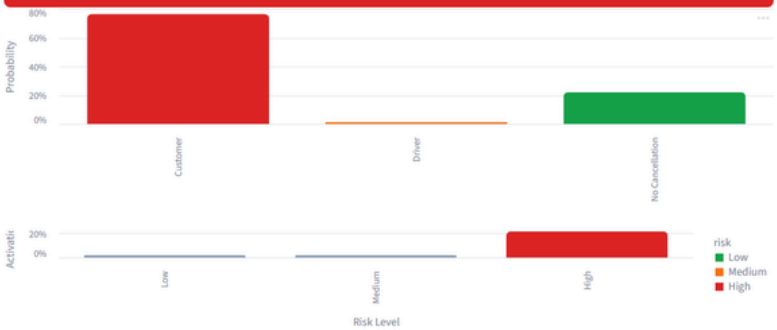
**Peak Hour time ranges:**

- Morning Peak: 07:00 - 10:00
- Evening Peak: 17:00 - 21:00
- Off-Peak: 10:00 - 17:00 / 21:00 - 07:00

## Real-time Cancellation Risk

| Customer Cancellation | Driver Cancellation | No Cancellation |
|---|---|---|
| 76.5% | 1.4% | 22.1% |

**High RISK**







> Input summary

> Model Insights and Notes

## AI Operational Recommendations

- URGENT: Increase driver incentives immediately.
- Reassign nearby drivers to high-risk areas.
- Traffic heavy — prioritize nearby drivers and avoid long-distance assignments.
- Driver ratings low — monitor driver behavior and quality.